

Trusst-It!

(Let's Bring out the Civil Engineer in us and Build a Simple Truss Bridge)

I. Some Funda Behind Bridges

You might have seen these bridges at many places



Wondered the engineering behind it? It's quite interesting & easy to Grasp
Watch these two videos and to know the secret sauce that civil & structural engineers use to build these bridges.

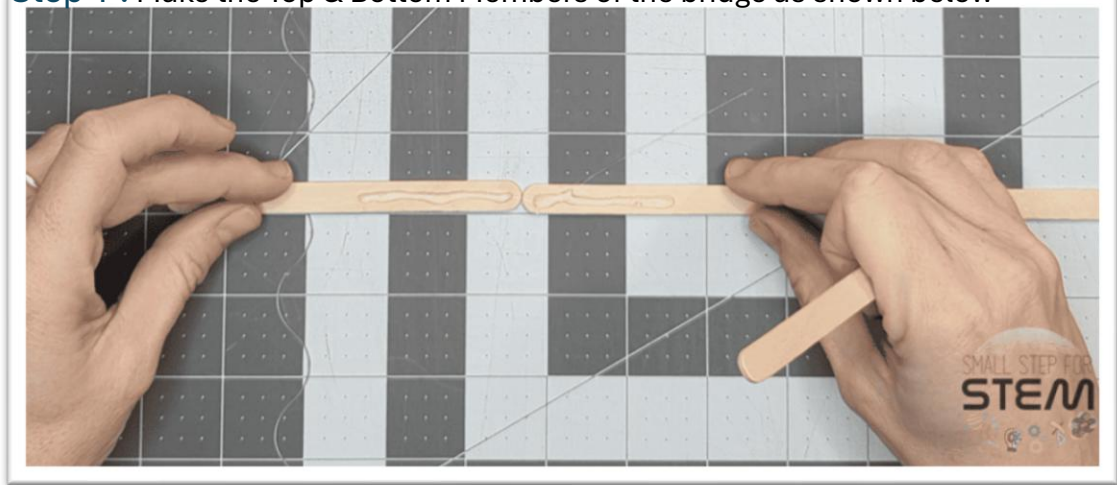
1. [Why bridges have triangular shapes?](#)
2. [Why are Triangles Stronger than Squares?](#)

II. Let's Build a Mini version Truss Bridge (Physical)

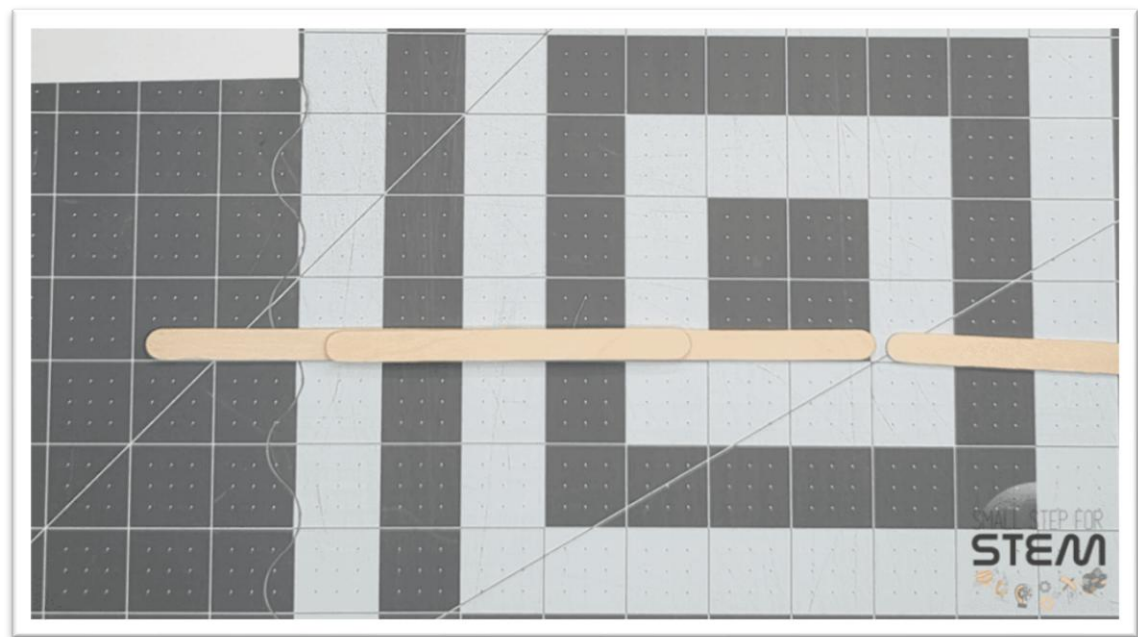
Estimated time : 30-45 mins as team activity

(Courtesy : [Small steps for STEM](#)) .

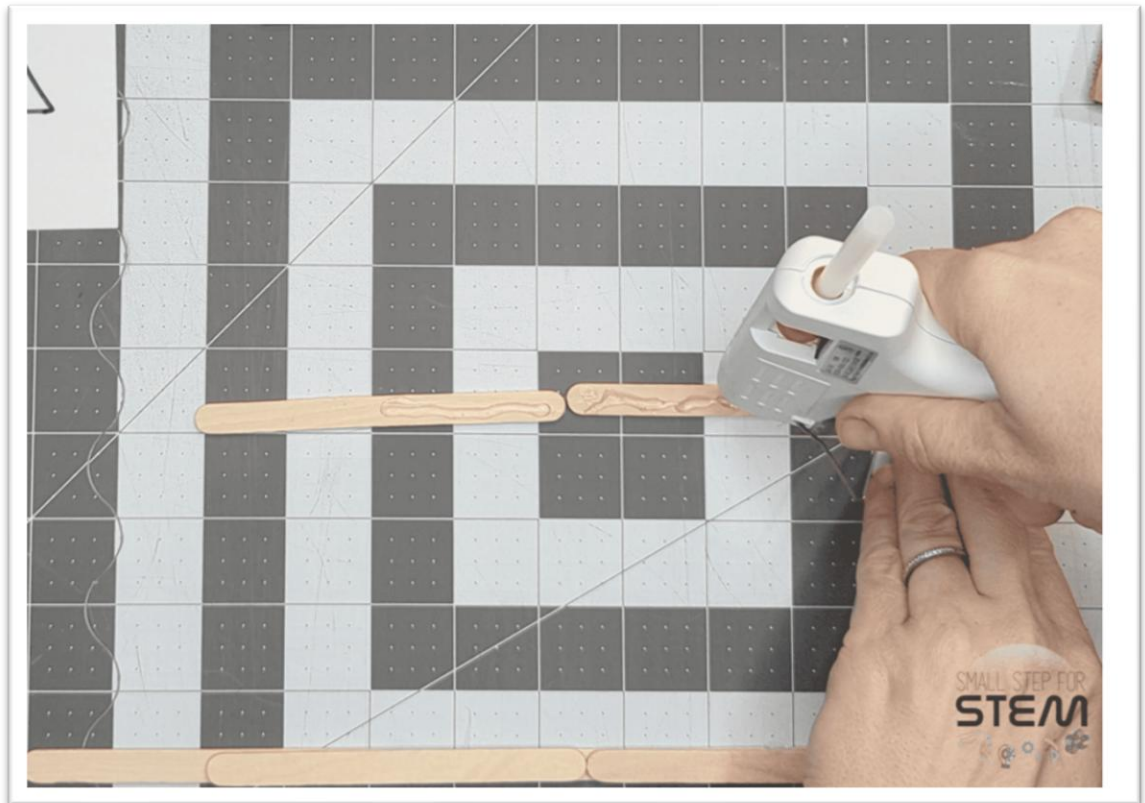
Step 1 : Make the Top & Bottom Members of the bridge as shown below



Glue applied at the edges as shown

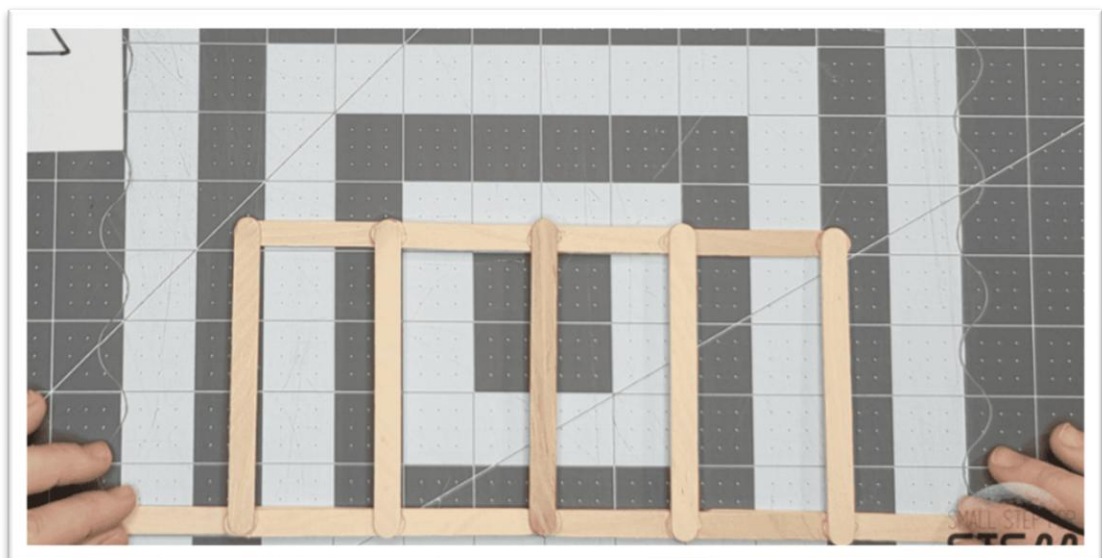


Both Sticks are Joined and make another joint with one more stick
(3 base sticks with 2 connecting sticks)



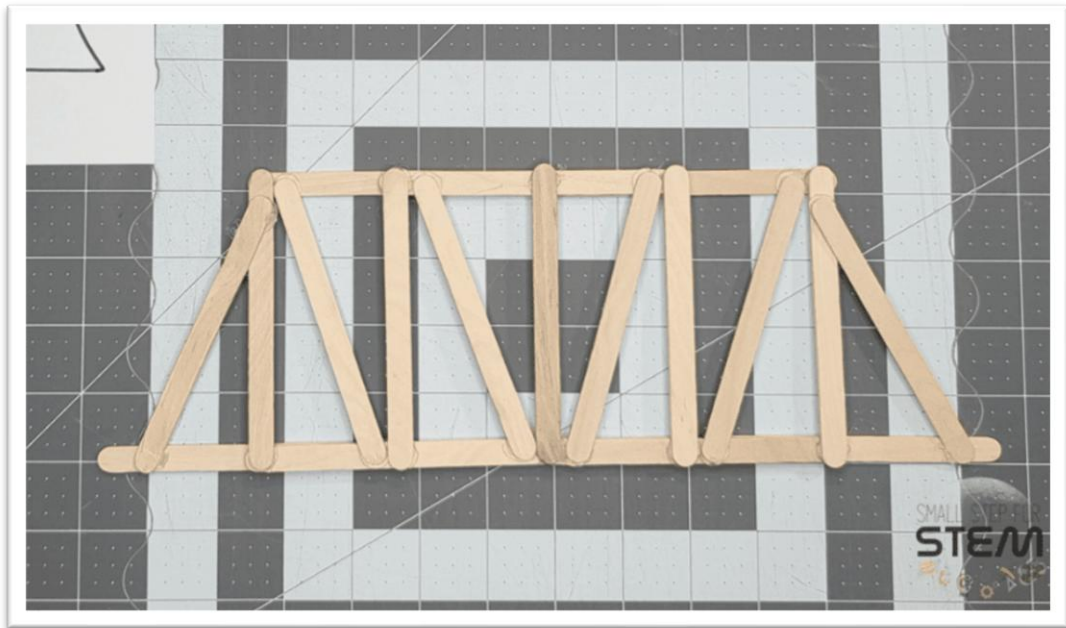
Follow the same approach for Top Beam (with one less stick)
(2 base stick with one connecting stick)

Step 2 : Connect the Top and Bottom Beam with Vertical Members



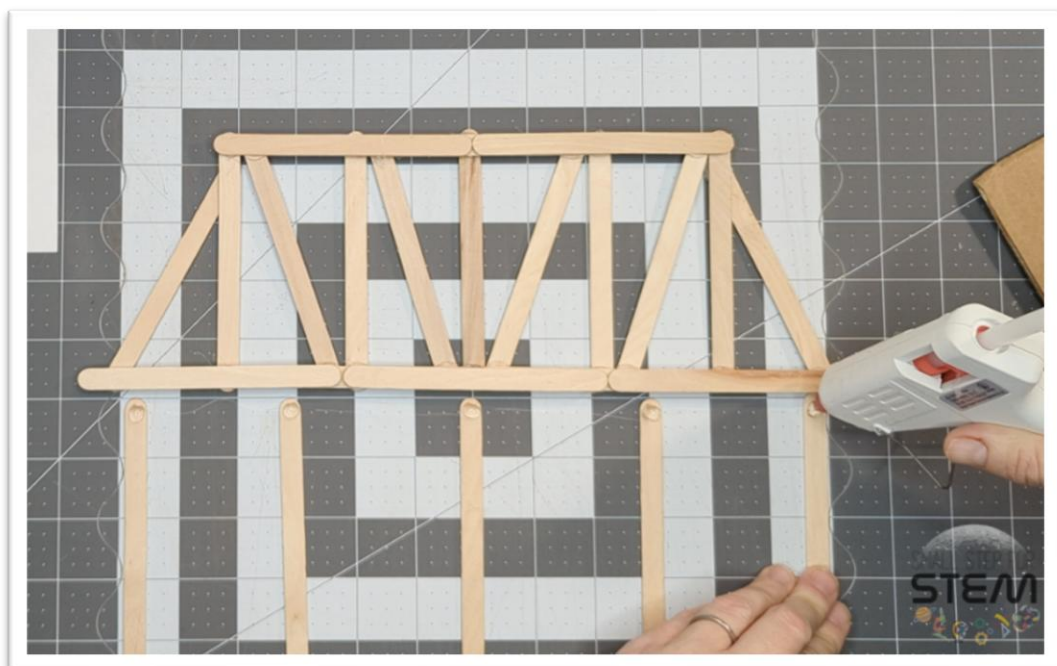
(Total 5 Vertical Members as shown)

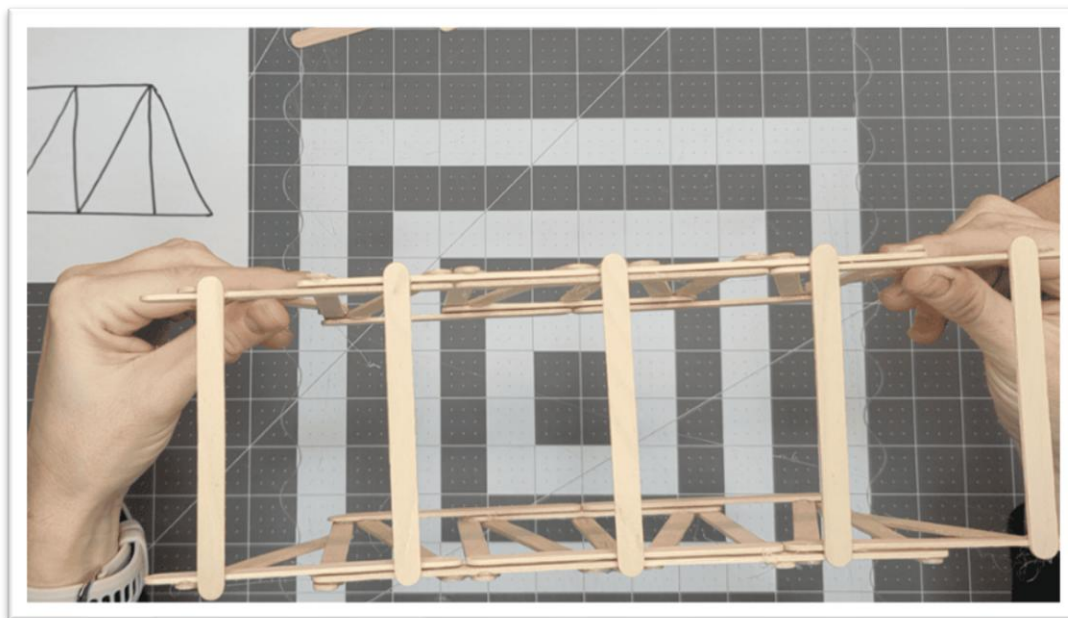
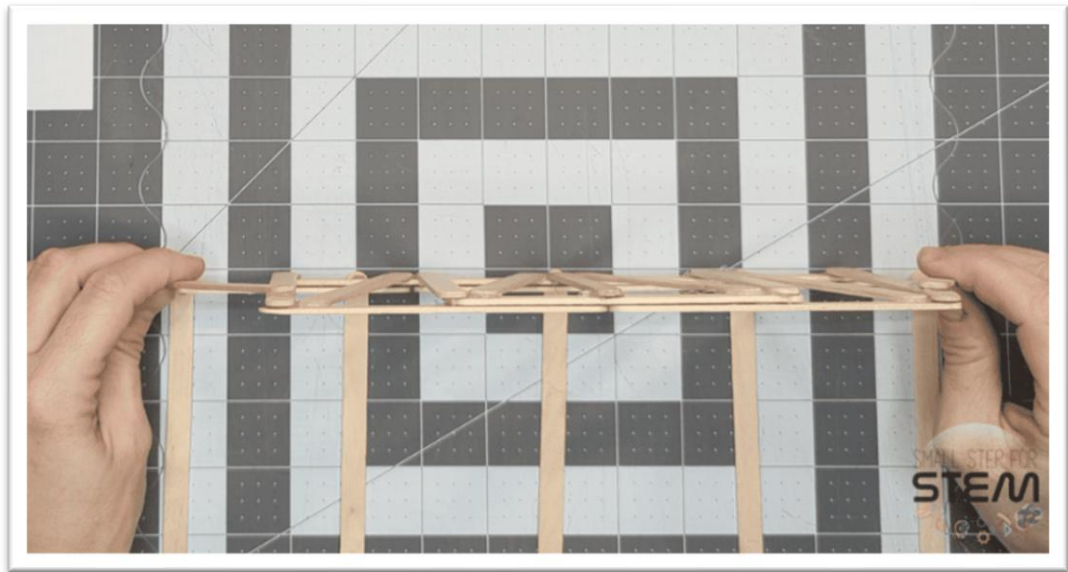
Step 3 : Make the Angle Members as shown

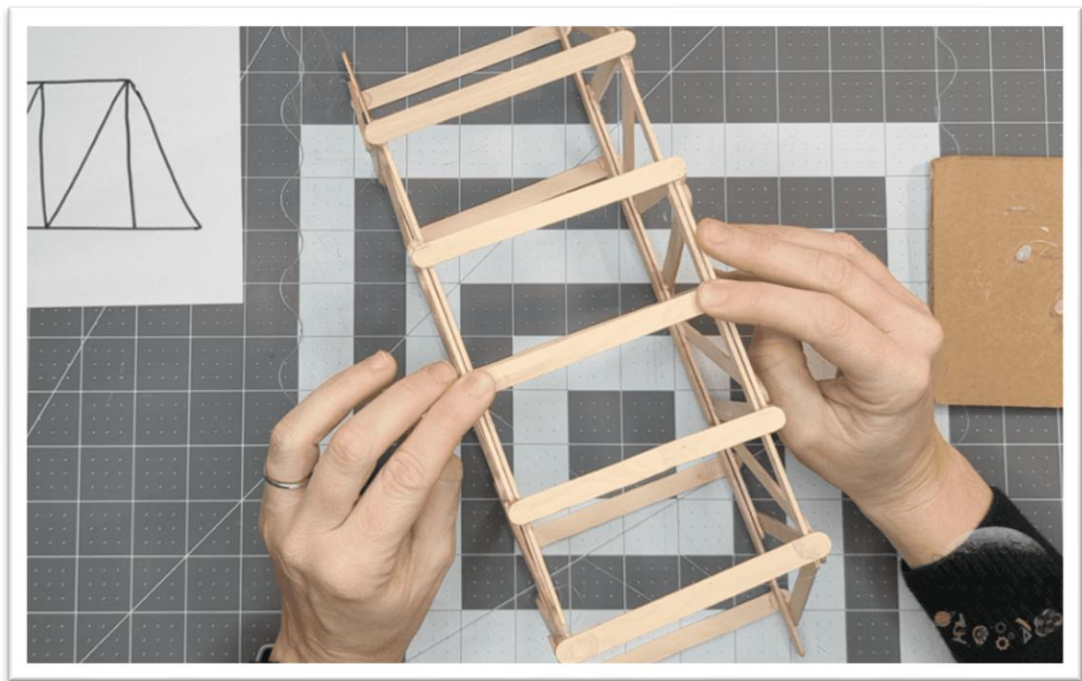


Step 4: Repeat Steps 1-3 and make the other side of the bridge

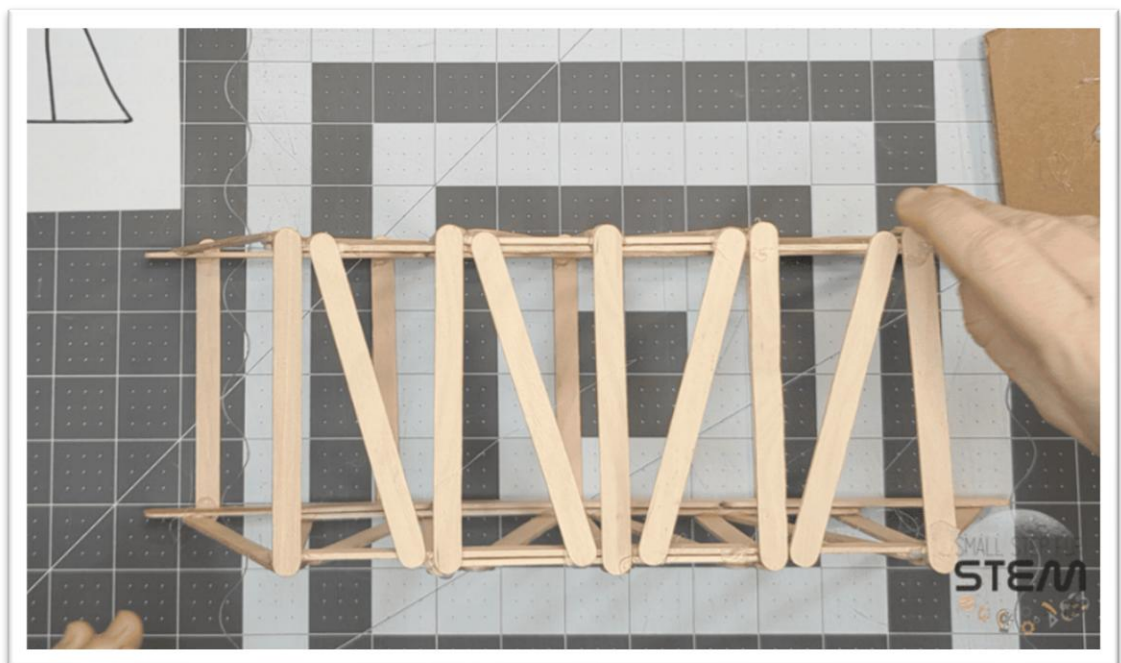
Step 5: Connect the two sides as shown





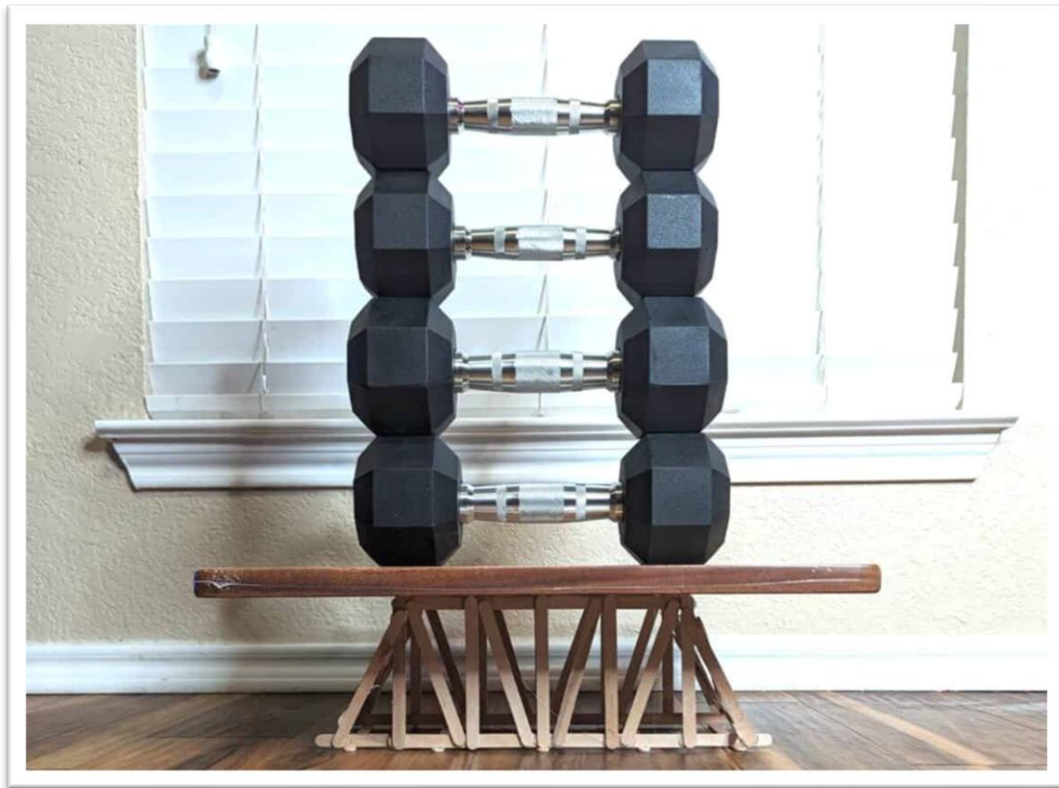


Step 6 : Add Angled Support for top side as shown



III. Time for Testing

Load the bridge from top slowly and see the fun unload!



IV: Know the Engineering details

To know more about the stress developed in each member when load is applied , play with this truss bridge simulator ([John Hopkins Simulator](#)) and have fun

[Intro Video](#) to get started.